

Airbnb NYC 2019

Data Storytelling & Statistical Validation

ApexPlanet Internship – Task 4

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Project Objective

- Analyze Airbnb NYC 2019 listings.
- Identify pricing and availability patterns.
- Understand room type distribution.
- Generate business insights through data analysis.

Data Cleaning & Preparation

- Missing values identified and handled.
- Duplicate records checked.
- Outliers analyzed and treated.
- New categories created:
 - Price Category
 - Availability Category
 - Review Activity

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	name	host_id	host_name	neighbourhood	neighbourhood	latitude	longitude	room_type	price	minimum	number_of_reviews	last_review	reviews_per_month	calculated_host_cancellation_rate	availability_365	
2	Furnished room in a historic building	20582832	Kathrine	Queens	Astoria	40.7681	-73.91651	Private room	10000	100	2	13/02/16	0.04		1	0
3	1-BR Lincoln Square	72390391	Jelena	Manhattan	Upper West Side	40.77213	-73.98665	Entire home/apt	10000	30	0				1	83
4	Luxury 1 bed in the heart of Manhattan	5143901	Erin	Brooklyn	Greenpoint	40.7326	-73.95739	Entire home/apt	10000	5	5	27/07/17	0.16		1	0
5	Quiet, Clean, Cozy 1BR in Manhattan	3906464	Amy	Manhattan	Lower East Side	40.71355	-73.98507	Private room	9999	99	6	01/01/16	0.14		1	83
6	2br - The Heart of Manhattan	4382127	Matt	Manhattan	Lower East Side	40.7198	-73.98566	Entire home/apt	9999	30	0				1	365
7	Spanish Harlem	1235070	Olson	Manhattan	East Harlem	40.79264	-73.93898	Entire home/apt	9999	5	1	02/01/15	0.02		1	0
8	Beautiful/Spanish Harlem	18128455	Rum	Manhattan	Tribeca	40.72197	-74.00633	Entire home/apt	8500	30	2	18/09/18	0.18		1	251
9	Film Location in the heart of Manhattan	1177497	Jessica	Brooklyn	Clinton Hill	40.69137	-73.96723	Entire home/apt	8000	1	1	15/09/16	0.03		11	365
10	East 72nd St Cozy 1BR in Manhattan	156158778	Sally	Manhattan	Upper East Side	40.76824	-73.95989	Entire home/apt	7703	1	0				12	146
11	Gem of east Manhattan	1262534951	Sandra	Brooklyn	East Flatbush	40.65724	-73.9245	Private room	7500	1	8	07/07/19	6.15		2	179
12	70' Luxury Midtown	7407743	Jack	Manhattan	Battery Park City	40.71162	-74.01693	Entire home/apt	7500	1	0				1	364
13	3000 sq ft duplex in the heart of Manhattan	3750764	Kevin	Manhattan	Chelsea	40.7506	-74.00388	Entire home/apt	6800	1	0				6	364
14	Luxury Tribeca	271248669	Jenny	Manhattan	Tribeca	40.71206	-74.00999	Entire home/apt	6500	180	0				1	365
15	Apartment in the heart of New York City															
16	Kitchens in the heart of New York City	35303743	Patricia	Manhattan	Upper West Side	40.76835	-73.98367	Private room	6500	30	0				1	97
17	SUPER BOWL in the heart of New York City	11598359	Jonathan	Brooklyn	Clinton Hill	40.68766	-73.96439	Entire home/apt	6500	1	0				1	0
18	Park Avenue	156158778	Sally	Manhattan	Upper East Side	40.78517	-73.9527	Entire home/apt	6419	1	0				12	45
19	UWS 1BR w/ kitchen in the heart of New York City	2151325	Jay And Liz	Manhattan	Upper West Side	40.77782	-73.97848	Entire home/apt	6000	14	17	17/02/15	0.27		1	359
20	Luxury townhouse in the heart of New York City	66240032	Linda	Manhattan	Greenwich Village	40.73046	-73.99562	Entire home/apt	6000	1	0				1	0
21	SuperBowl Party in the heart of New York City	1483320	Omri	Manhattan	Little Italy	40.71895	-73.99793	Entire home/apt	5250	1	0				1	0
22	Midtown Manhattan	16105313	Debra	Manhattan	Midtown	40.74482	-73.98367	Entire home/apt	5100	30	1	22/06/19	1		2	343
23	Near Williamsburg	14908606	Bianca	Brooklyn	Bedford-Stuyvesant	40.69572	-73.95731	Private room	5000	6	10	02/01/16	0.16		1	363
24	Broadway 1	153497815	Sarah-B	Brooklyn	Bedford-Stuyvesant	40.68742	-73.91957	Entire home/apt	5000	2	8	09/12/17	0.38		1	0
25	Fulton 2	100069033	Sarah-2	Brooklyn	Cypress Hills	40.68185	-73.88128	Entire home/apt	5000	2	4	03/01/18	0.18		1	0
26	Beautiful 1 BR in the heart of New York City	213266	Jessica	Manhattan	NoLita	40.72193	-73.99379	Entire home/apt	5000	1	2	28/09/13	0.03		1	365
27	4-Floor Uniq in the heart of New York City	172611460	Rasmus	Manhattan	Harlem	40.82511	-73.94961	Entire home/apt	5000	1	2	23/02/19	0.38		2	150
28	Victorian Film in the heart of New York City	2675644	Alissa	Staten Island	Randall's Manor	40.63952	-74.0973	Entire home/apt	5000	1	0				1	344
29	Northside Williamsburg	956324	Alex	Brooklyn	Williamsburg	40.71705	-73.9647	Entire home/apt	4500	30	0				1	365
30	Photography in the heart of New York City	1177497	Jessica	Brooklyn	Clinton Hill	40.69127	-73.96563	Entire home/apt	4500	1	5	29/12/18	0.09		11	365
31	Beautiful private room in the heart of New York City	8748976	Jeffrey	Brooklyn	Bedford-Stuyvesant	40.68807	-73.95426	Private room	4200	114	0				1	347
32	bay ridge & stuyvesant	263564234	Nony	Brooklyn	Bay Ridge	40.63087	-74.02006	Entire home/apt	4200	60	0				1	90
33	Chelsea Gall in the heart of New York City	3750764	Kevin	Manhattan	Chelsea	40.74888	-74.00481	Entire home/apt	4160	1	0				6	364
34	30 days mini in the heart of New York City	177396569	Yanina	Manhattan	Hell's Kitchen	40.76043	-73.99132	Entire home/apt	4100	30	0				3	180
35	Sunny, Family in the heart of New York City	4289240	Lucy	Brooklyn	Prospect Heights	40.67919	-73.97191	Entire home/apt	4000	4	0				1	83
36	NYC SuperBlock in the heart of New York City	11353904	Todd	Manhattan	Upper West Side	40.79476	-73.97299	Entire home/apt	4000	1	0				1	0
37	The Otheroom in the heart of New York City	18037301	James	Manhattan	West Village	40.73571	-74.0078	Entire home/apt	4000	1	0				1	173
38	Greenwich Village	147814925	Lloyd	Manhattan	Greenwich Village	40.733	-73.99413	Entire home/apt	3900	30	7	13/03/18	0.32		1	180
39	VIP Duplex Townhouse in the heart of New York City	49186997	Joseph	Manhattan	Greenwich Village	40.73305	-73.99373	Entire home/apt	3800	30	2	27/01/17	0.06		1	180
40	Prime NYC Location in the heart of New York City	1427243	Jordana	Manhattan	East Village	40.73323	-73.98859	Entire home/apt	3750	1	0				1	0
41	A Night at the Museum in the heart of New York City	45863742	James	Manhattan	Battery Park City	40.71364	-74.01758	Entire home/apt	3750	1	0				1	365
42	West 84th St in the heart of New York City	156158778	Sally	Manhattan	Upper West Side	40.78545	-73.97123	Entire home/apt	3613	1	1	15/06/19	1		12	158
43	NYC HOLIDAY in the heart of New York City	2039658	Pam	Manhattan	Upper East Side	40.76519	-73.96874	Entire home/apt	3600	31	0				1	358
44	East 7th Street in the heart of New York City	156158778	Sally	Manhattan	East Village	40.72779	-73.98644	Entire home/apt	3518	1	0				12	215
45	West 88th St in the heart of New York City	156158778	Sally	Manhattan	Upper West Side	40.78742	-73.97011	Entire home/apt	3512	1	0				12	124
46	Chelsea Gall in the heart of New York City	3750764	Kevin	Manhattan	Chelsea	40.74913	-74.00373	Private room	3210	1	0				6	364
47	Furnished 10 in the heart of New York City	229458601	Kay	Manhattan	Harlem	40.81315	-73.9511	Entire home/apt	3200	90	0				1	365
48	Breath-taking in the heart of New York City	836168	Henry	Manhattan	Upper West Side	40.7791	-73.98182	Entire home/apt	3000	30	8	10/12/16	0.24		11	365
49	SPECTACULAR in the heart of New York City	6145729	Stephanie	Manhattan	SoHo	40.72605	-74.00572	Entire home/apt	3000	7	1	30/06/19	1		1	325
50	Beautiful 3 br in the heart of New York City	256239	Tracey	Manhattan	Upper West Side	40.80142	-73.96931	Private room	3000	7	0				1	365
51	CQ 1 Bedroom in the heart of New York City	30223355	Adrienne	Manhattan	Financial District	40.70706	-74.00991	Private room	3000	3	0				1	23
52	Massive Loft in the heart of New York City	51340124	Nikki	Manhattan	Chelsea	40.73851	-73.99213	Entire home/apt	3000	2	0				2	0
53	Luxury West in the heart of New York City	101080203	Luxury Proper	Manhattan	Gramercy	40.73652	-73.98618	Entire home/apt	3000	1	1	16/12/18	0.15		1	365

Exploratory Data Analysis

Insert:

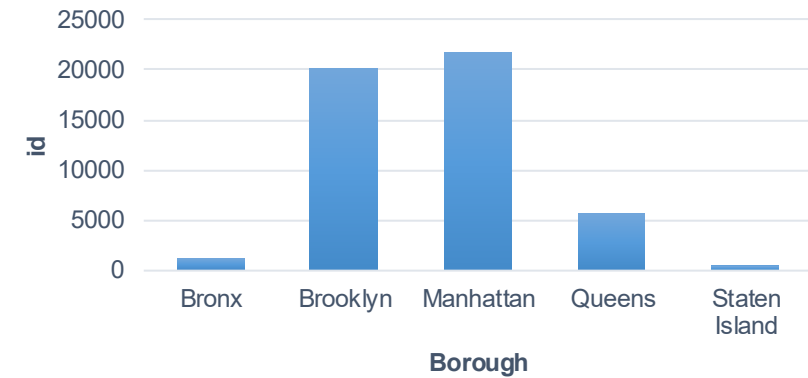
Listings by Borough chart

Room Type Distribution chart

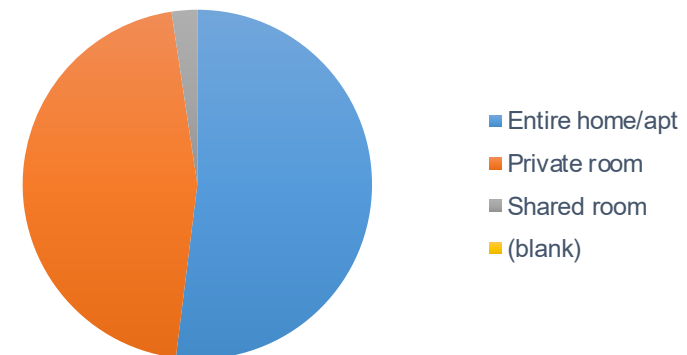
Key Findings

1. Manhattan has the highest number of listings.
2. Entire Home/Apt is the most common room type.

Airbnb listings by Borough



Distribution of Airbnb Room type



Business Intelligence Analysis

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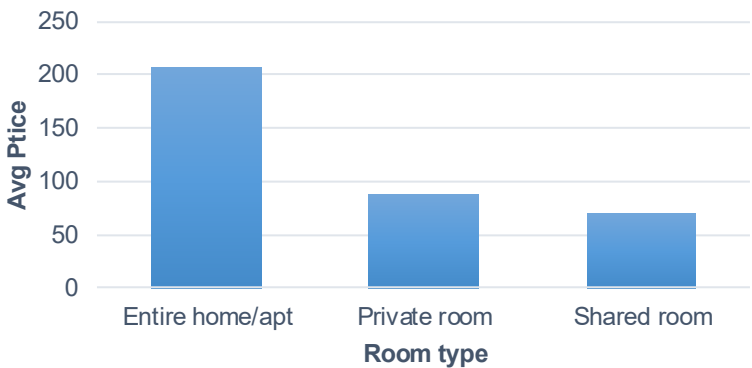
Average Price by Room Type

Average Availability by Borough

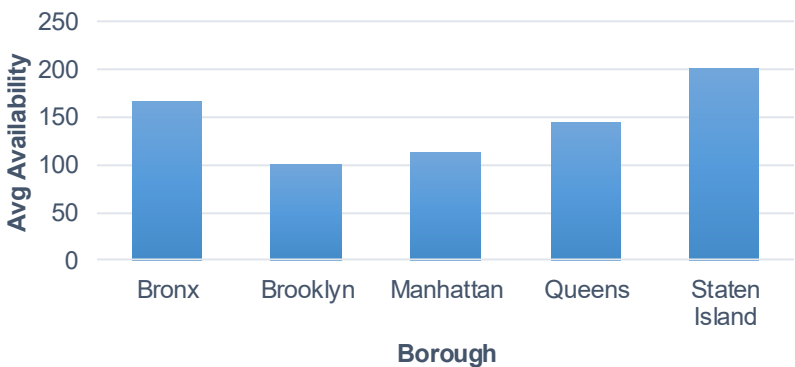
Key Findings

- Entire Home/Apt has the highest average price.
- Availability varies across boroughs.

Average Price by Room Type



Average Availability by Borough



Correlation Analysis

Price vs Availability Analysis.

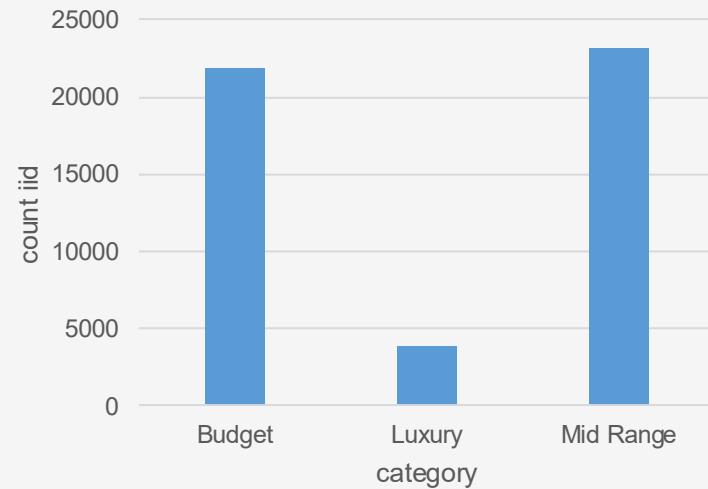


Finding

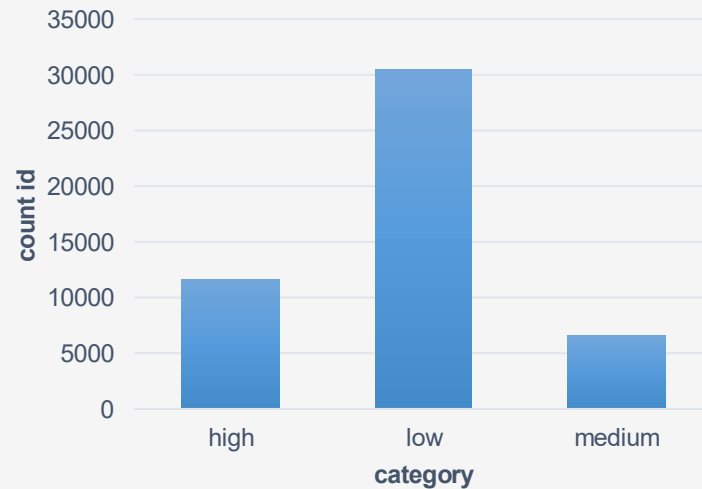
A weak relationship exists between price and availability.

Deep Dive Analysis

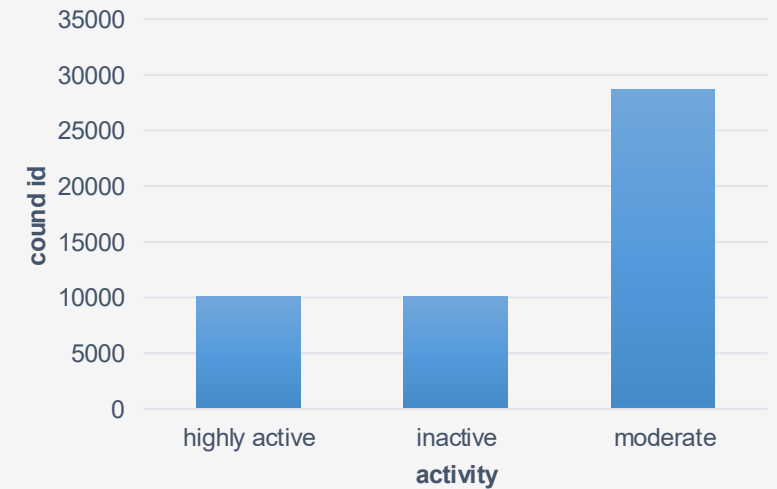
Listing Distribution by price category



Listing Distribution by availability category



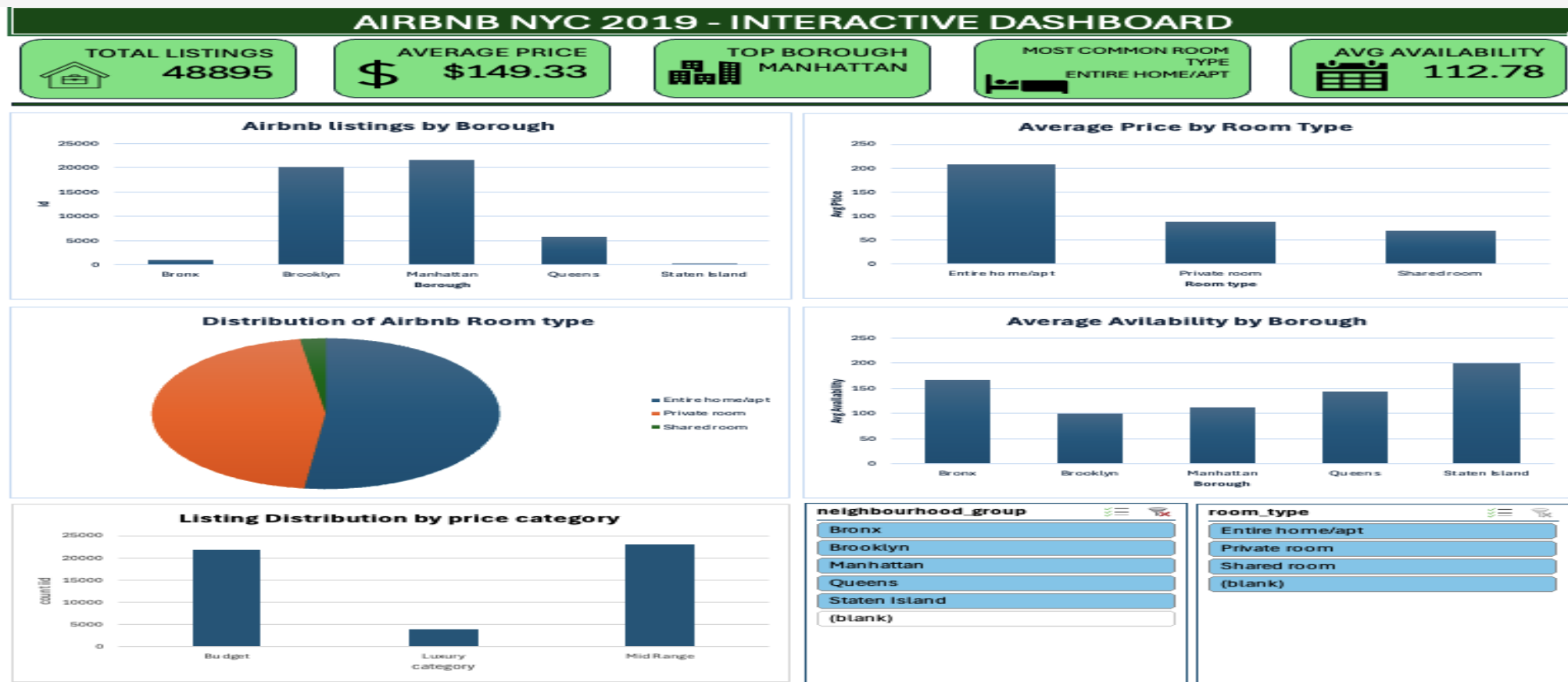
Listing Distribution by review activity



FINDING

Segmentation reveals different listing patterns and customer engagement levels.

Interactive Dashboard



Hypothesis Testing

H_0 (Null Hypothesis):

Average price is the same across all room types.

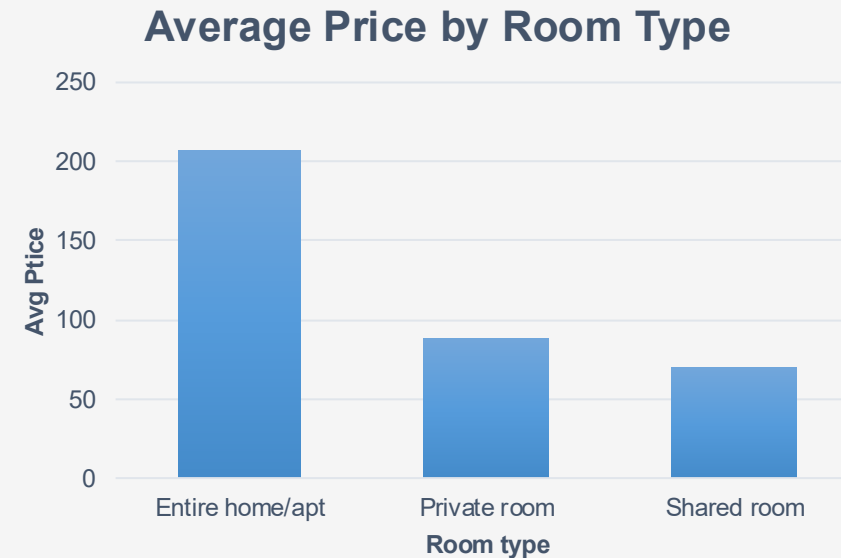
H_1 (Alternative Hypothesis):

Average price differs across room types.

Room Type	Average Price
Entire Home/Apt	207.73
Private Room	86.28
Shared Room	70.13

Conclusion

The average price differs significantly across room types. Therefore, room type influences listing price.



Recommendations & Conclusion

Recommendations

- Focus on high-performing boroughs.
- Optimize pricing based on room type.
- Improve occupancy strategies.
- Use segmentation for targeted decisions.

Conclusion

The project demonstrates how data cleaning, EDA, dashboarding, and segmentation can support data-driven business decisions.